Graph Signal Processing and Interpolation for EEG Channel Reconstruction

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***Abstract*—This paper studies EEG channel reconstruction under missing-channel conditions using a graph signal processing (GSP) benchmark with reproducible execution artifacts. Following the GSP perspective popularized by Ortega and collaborators, we model EEG electrodes as graph vertices and evaluate reconstruction quality through both spatial and temporal criteria. We compare geometric baselines, graph-based estimators, and graph-time regularization methods under scenario-driven masking protocols. The current B2 stage includes full-scale batched runs and consolidated publication tables with MAE, RMSE, DTW, and SNR statistics. Results indicate that graph-aware and TV/time methods remain consistently competitive, while ranking differences are scenario-dependent and therefore require variance-aware interpretation. We also report the current BGSRP status transparently: strong proxy performance in Python, with strict MATLAB/GSPBox 1:1 equivalence still pending.**

***Index Terms*—EEG reconstruction, graph signal processing, interpolation, missing channels, temporal regularization**

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